

| Question |     |  |  | Marking details                         | Marks available |     |     |       |       |      |
|----------|-----|--|--|-----------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) |  |  | its density is 1.7 g/cm <sup>3</sup>    |                 |     | 1   | 1     |       | 1    |
|          |     |  |  | its melting point is 650 °C             |                 |     |     |       |       |      |
|          |     |  |  | it fizzes vigorously with sulfuric acid |                 |     |     |       |       |      |
|          |     |  |  | it is malleable                         |                 |     |     |       |       |      |
|          | (b) |  |  | it does not react with sulfuric acid    |                 |     | 1   | 1     |       |      |
|          |     |  |  | it is ductile                           |                 |     |     |       |       |      |
|          |     |  |  | it would melt when it lands on planet J |                 |     |     |       |       |      |
|          |     |  |  | its density is 11.3 g/cm <sup>3</sup>   |                 |     |     |       |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) |  |  | award (1) for each correct answer<br><br>it does not react with sulfuric acid <input checked="" type="checkbox"/><br><br>it is expensive <input type="checkbox"/><br><br>it is a good conductor of heat <input type="checkbox"/><br><br>it is non-magnetic <input type="checkbox"/><br><br>it has a melting point much higher than the temperature on planet J <input checked="" type="checkbox"/><br><br>it is shiny so will reflect the sun's rays <input type="checkbox"/><br><br>if three boxes ticked each incorrect answer negates a correct one<br>0 if four or more boxes ticked |                 |     | 2   | 2     |       |      |
|          | (d) |  |  | <b>2Na + H<sub>2</sub>SO<sub>4</sub> → Na<sub>2</sub>SO<sub>4</sub> + H<sub>2</sub></b><br><br>award (1) for correct formula Na <sub>2</sub> SO <sub>4</sub><br><br>award (1) for balancing <b>only</b> if Na <sub>2</sub> SO <sub>4</sub> given                                                                                                                                                                                                                                                                                                                                         |                 | 2   |     | 2     | 1     |      |
|          |     |  |  | Question 5 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0               | 2   | 4   | 6     | 1     | 1    |